AP CALCULUS AB

Unit 1 • Lesson 1.13 practice

# Lesson 1.13 • Removing Discontinuities

5 multiple-choice questions and 1 free-response question

**Directions**

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

## Question 1

NO CALCULATOR

Let $g(x) = \dfrac{x^2 - 16}{x - 4}$. What value of $g(4)$ makes $g$ continuous at $x = 4$?

(A) 0
(B) 4
(C) 16
(D) 8

## Question 2

NO CALCULATOR

For what value of $k$ is $f(x) = \begin{cases} kx^2, & x \leq 1 \\ x + 3, & x > 1 \end{cases}$ continuous?

(A) 4
(B) 3
(C) 1
(D) 2

## Question 3

NO CALCULATOR

For what value of $a$ is $f(x) = \begin{cases} ax + 2, & x < 3 \\ x^2 - a, & x \geq 3 \end{cases}$ continuous?

(A) $\dfrac{11}{4}$
(B) $\dfrac{7}{4}$
(C) $\dfrac{7}{2}$
(D) $7$

## Question 4

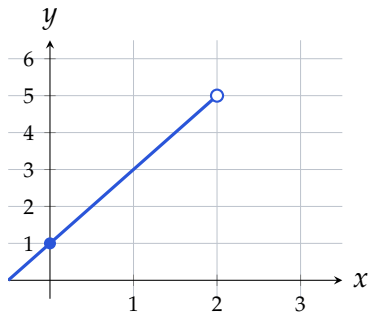
NO CALCULATOR

Let $f(x) = \dfrac{\sin x}{x}$ for $x \neq 0$. What value of $f(0)$ makes $f$ continuous at 0?

(A) 0
(B) undefined
(C) 1
(D) $\pi$

**Question 5** (graph) NO CALCULATOR

For $x < 2$, the graph of $f$ is the line shown, through $(0,1)$ and $(2,5)$. For $x \geq 2$, $f(x) = x^2 + k$. For what value of $k$ is $f$ continuous at $x = 2$?




(A) 5
(B) $-1$
(C) 9
(D) 1

**Question 6** (Free response) NO CALCULATOR

Let $f(x) = \begin{cases} 2x + a, & x < 1 \\ bx^2 + 1, & x \geq 1 \end{cases}$ where $a$ and $b$ are constants, and $f(2) = 9$.

(a) Find $b$.
(b) Find the value of $a$ that makes $f$ continuous at $x = 1$.
(c) With these values, verify that $f$ is continuous at $x = 1$ using the definition.